

MuLERMOCs — Organic Nomenclature Learning Platform

A Greek-first interactive learning platform for organic nomenclature, built around molecular structure, naming logic, and multiple representations.

Current working baseline: v39

Main entry point: [mulermoc-nom-39.html](#)

Core engine: [js/mulermoc-nom-core-39.js](#)

Viewer layer: [js/mulermoc-nom-molview-39.js](#)

Teaching layer: [js/mulermoc-nom-teaching-39.js](#)

Data set: [js/jsme-nick-nomenclature-moc2-data_39.js](#)

Project status

This project is currently a no-build web application built with JSME, JSmol, Snap.svg, and vanilla JavaScript. It already contains a working nomenclature engine and multiple representation modes, but it is still a legacy HTML-based implementation rather than a fully refactored product structure.

The next product iteration is being re-scoped around a clearer educational model:

- Greek-first interface
 - homologous series as the primary navigation model
 - rules as a secondary explanatory layer
 - chemistry expansion in this order: branched molecules → ethers → esters
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Current stack

- **jQuery** 3.7.1
 - **JSME** — 2D molecule editor/viewer
 - **JSmol** — 3D molecule viewer
 - **Snap.svg** — SVG manipulation
 - **Web Speech API** — Greek TTS narration
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What exists now

Supported baseline

- 2D molecular rendering via JSME
- 3D molecular rendering via JSmol
- Multiple representation modes:
 - condensed
 - expanded
 - skeletal

- annotated skeletal
- Greek IUPAC name generation
- Name analysis and explanation panels
- Rule theory panel
- TTS narration in Greek
- PNG export for 2D and 3D views
- Responsive menu and settings controls
- Current molecule data set for the teaching application

Current implementation architecture

The project uses a split-layer architecture:

- **Core logic:** analysis + naming
- **Viewer layer:** JSME / JSmol / Snap.svg integration
- **Teaching layer:** rule explanations, narration, UI wiring
- **Data layer:** molecule definitions and structural data

This makes the codebase a workable foundation for the restart and expansion.

Restart and expansion plan

Product principle

The next version should not be organized primarily around rule-by-rule teaching. Instead, the platform should be organized around:

- homologous series
- molecule browsing by family
- molecule detail view
- naming analysis
- optional rule explanations

Immediate chemistry roadmap

1. Branched molecules

- alkyl substituent recognition
- multi-branch chain selection
- corrected locant logic
- validation examples

2. Ethers

- detection and classification
- naming logic
- examples and edge cases

3. Esters

- acid-derived and alcohol-derived grouping
- full IUPAC naming logic
- validation dataset

After the chemistry expansion

- add a first learning interaction layer
- add atom/bond click targets
- add one exploration mode
- then consider quiz features and research instrumentation

Deferred scope

These remain explicitly out of the first restart milestone:

- full research-mode architecture
- participant/session logging
- quiz SPA
- condition locking
- CSV export
- transfer-test data set
- server-side classroom management
- rings and aromatics
- full reaction module

These belong to a later product phase.

Key project documents

- [DEVELOPMENT-PLAN.md](#)
- [RESEARCH-PLAN.md](#)
- [PROJECT-PLAN.md](#)
- [PROJECT-PLAN-GR.md](#)
- [CHANGELOG.md](#)

These plans describe the research and product ambitions, but the current implementation should be treated as the working v39 baseline rather than as a fully finished research platform.

Technical notes

Important observations

- The workspace contains multiple historical versions, including v35-v39 snapshots.
- The project plans still describe older intended milestones and should be reconciled with the real v39 baseline.
- The chemistry engine should be validated before expanding into more complex categories.

- Known engine issues still need explicit testing, especially around ester and tertiary-amine detection paths.
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Future goal

The target product is not only a naming tool, but a series-based chemistry learning environment that helps students explore:

- molecular family
- naming patterns
- structural logic
- representation transitions
- explanation by structure, not only by rules

This makes the platform more pedagogically coherent and more scalable than a strict rule-first interface.

Repository status summary

- Working baseline: **v39**
- Main focus: **stable learning app + chemistry expansion**
- Near-term goal: **branch logic, ethers, esters**
- Product model: **homologous series first**
- Rules: **secondary explanatory layer**
- Research layer: **deferred**